

If you don't have a model - use for the magma strip

Name: _____
Mrs. King
Integrated Sci. I, Period _____
Date: _____

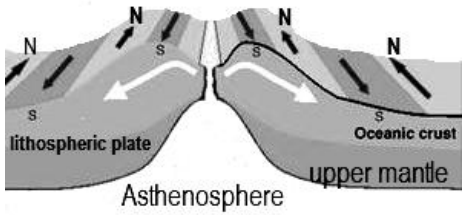
Assignment #52: Sea Floor Spreading Lab

§ BACKGROUND

In 1963, two English geophysicists, Vine and Matthews, were using extremely sensitive magnetometers to make measurements of the seafloor across the Mid-Atlantic Ridge. They already knew that over the 4.6 billion years of Earth's history, its magnetic poles have changed directions more than once, so that the South Pole and the North Pole have actually switched polarity. their measurement showed a pattern of magnetic reversals around the ridge.

When a volcano erupts, the molten rocks have no particular polarity. However, as they cool, the rocks become magnetically aligned with the earth's magnetic poles like a compass. The rocks record the direction of the poles at that time: the "rock compass pointing" North when the poles are "normal" and south when the poles are reversed.

Vine and Matthews discovered that the ages of the rocks aligned with the pole reversals. Also the rocks on both sides were progressively older as they moved away from the ridge.. This s a pattern of polar reversal stripes on both sides of the ridge showed that the rocks were formed at different times!



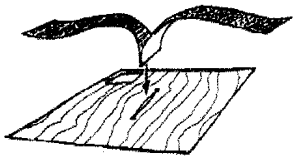
§ PURPOSE

Observe how magnetic reversal show seafloor is growing

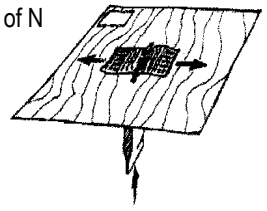
§ PRELAB:

Make a profile of the mid-ocean ridge that will be used in your sea floor spreading model. You will use the undersea topo map, called a bathmetric map on the inside page.

§ PROCEDURE



1. Get a ready-made model. If you don't have one, make your own model. Cut a along the slit line on the bathmetric topo map on the inside page. Next copy the magna strip on the opposite page onto another paper and cut it out. Continue to follow the rest of the directions.



2. On the model, the cardboard with the contour lines represents the mid-ocean ridge. The patterned band of paper with the N's and S's represents the magma. This magma strip has the same pattern of N and S on both sides of the side - the sides are mirror images!
3. Fold the patterned band on the fold line carefully insert the open end of the folded reversal pattern into the slit from underneath, holding onto the stub, and carefully pull the first several centimeters of paper through (up to the first dark line).
4. This represents a volcanic eruption during which the rocks adopt the magnetic field of the Earth at that time.

5. Continue to play with the pattern band, carefully pulling it through the vent. When you are done – answer the questions.

§ CONCLUSIONS/ QUESTIONS

1. What kind of plate boundary does this activity model?

2. What does each fold of paper represent as it is pulled up from below?

3. Where is the oldest volcanic rock located?

4. Where is the newest volcanic rock located?

5. Why do the pattern bands vary in width?

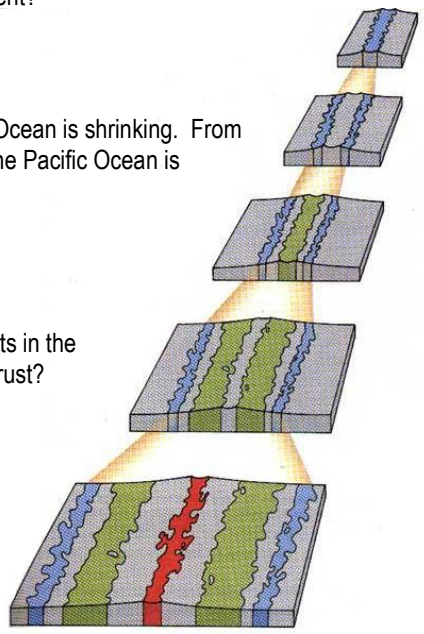
6. How does seafloor spreading provide evidence for plate tectonics?

7. What happens to the sea floor when it hits a continent?

8. While the Atlantic Ocean is expanding, the Pacific Ocean is shrinking. From what you have learned, how can you explain why the Pacific Ocean is shrinking?

9. What is the relationship between convection currents in the asthenosphere and the movement of the oceanic crust?

10. Explain what is going on in the picture to the right.



§ PRELAB:

Make a bathymetric profile of the mid-ocean ridge that will be used in your sea floor spreading model. Color the VENT RED; Color the RIDGE BROWN.

